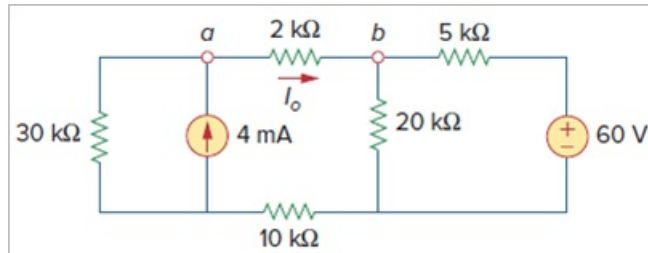


Problem

Consider the circuit in Fig. 4.144. An ammeter with internal resistance R_i is inserted between A and B to measure I_o . Determine the reading of the ammeter if: (a) $R_i = 500\ \Omega$, (b) $R_i = 0\ \Omega$. (Hint: Find the Thevenin equivalent circuit at terminals a-b.)

Figure 4.144:



Step-by-step solution

Step 1 of 9

To find R_{Th} , consider the circuit below,

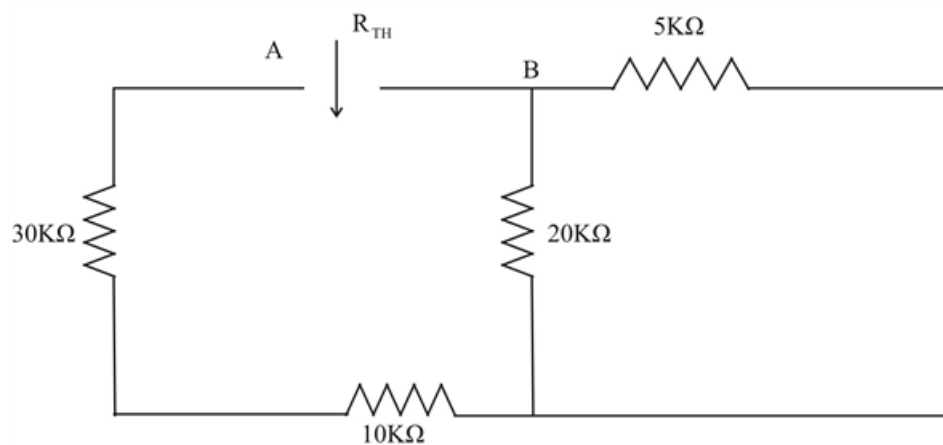


Fig-(a)

Step 2 of 9

From Fig-(a),

$$R_{Th} = (30\text{k}\Omega + 10\text{k}\Omega) + (20\text{k}\Omega \parallel 5\text{k}\Omega)$$

$$R_{Th} = 40 + \frac{20 \times 5}{20 + 5}$$

$$R_{Th} = 40 + \frac{100}{25}$$

$$R_{Th} = 40 + 4$$

$$R_{Th} = 44\text{k}\Omega$$

To find V_{Th} , consider the circuit below,

Step 3 of 9

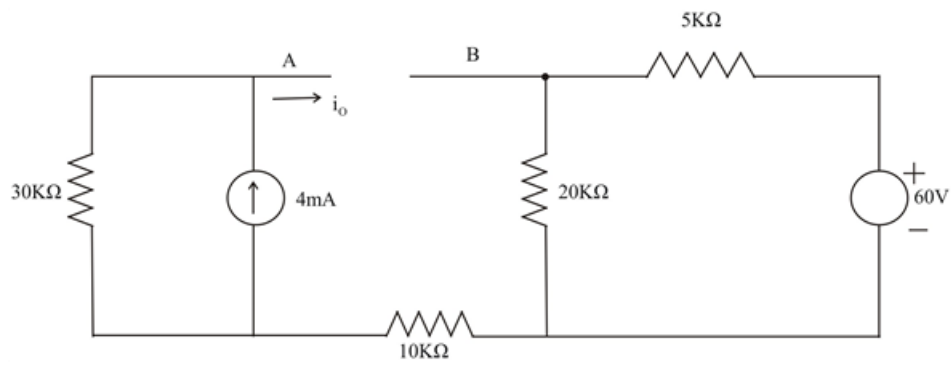


Fig-(b)

Step 4 of 9

From Fig-(b),

$$V_A = 30 \times 4$$

$$V_A = 120\text{V} \text{ and}$$

$$V_B = \left(\frac{20}{20+5} \right) \times (60)$$

$$V_B = \frac{20}{25} \times (60)$$

$$V_B = 48\text{V}$$

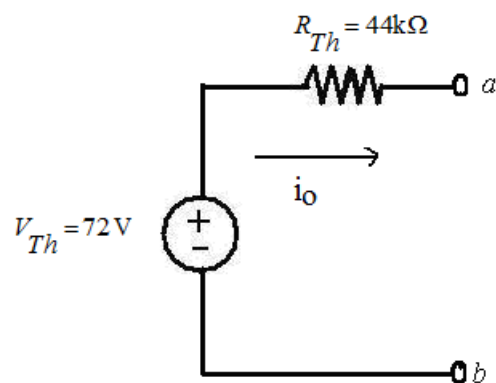
Therefore,

$$V_{Th} = V_A - V_B$$

$$V_{Th} = 72\text{V}$$

Step 5 of 9

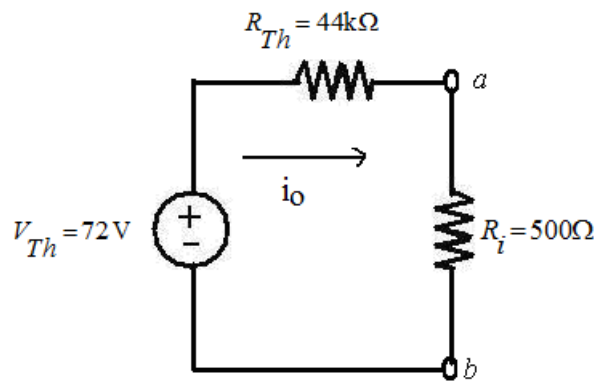
Therefore Thevenin's equivalent circuit as shown in below;



Step 6 of 9

(a)

$$R_i = 500\Omega$$



Step 7 of 9

Applying KVL to the above circuit we get,

$$-72 + (44 \times 10^3 + 500)I_o = 0$$

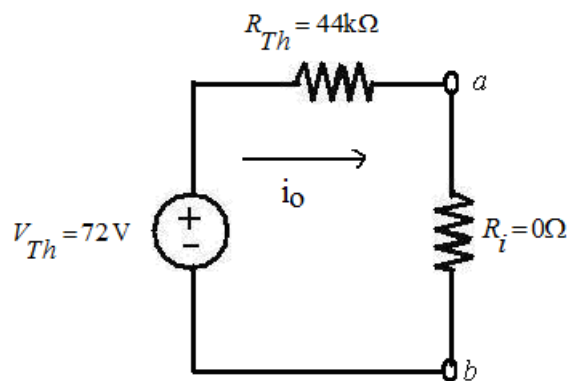
$$44500I_o = 72$$

$$i_o = 1.62\text{mA}$$

Step 8 of 9

(b)

$$R_i = 0\Omega$$



Step 9 of 9

Applying KVL to the above circuit we get,

$$-72 + (44 \times 10^3 + 0)I_o = 0$$

$$44000I_o = 72$$

$$i_o = 1.636\text{mA}$$